

Rozwiązania zadań z „Kółka matematycznego dla olimpijczyków” H.Pawłowskiego

Olaf Witusik
olaf.witusik@gmail.com

Luty, 2024

I SUMY I ILOCZYNY

Zadanie 8. Strony równania oznaczmy przez

$$\alpha(n) = \left(\sum_{i=1}^n a_i b_i \right)^2 \quad (1)$$

$$\beta(n) = \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right) - \sum_{1 \leq i < j \leq n} (a_i b_j - a_j b_i)^2 \quad (2)$$

T:. $\forall n \in \mathbb{N}_+ : \alpha(n) = \beta(n)$

D:. Dowód przez indukcję.

Dla $n = 1$:

$$\alpha(1) = a_1^2 b_1^2 = \beta(1) \quad (3)$$

Założmy, że dla $k \in \mathbb{N}_+$ zachodzi: $\alpha(k) = \beta(k)$. Tak więc:

$$\left(\sum_{i=1}^k a_i b_i \right)^2 = \left(\sum_{i=1}^k a_i^2 \right) \left(\sum_{i=1}^k b_i^2 \right) - \sum_{1 \leq i < j \leq k} (a_i b_j - a_j b_i)^2 \quad (4)$$

Wykażemy teraz, że z założenia indukcyjnego wynika prawdziwość tezy dla $k + 1$.

$$\alpha(k+1) = \left(\sum_{i=1}^{k+1} a_i b_i \right)^2 = \left(\sum_{i=1}^k a_i b_i + a_{k+1} b_{k+1} \right)^2 = \quad (5)$$

$$= \left(\sum_{i=1}^k a_i b_i \right)^2 + 2a_{k+1} b_{k+1} \sum_{i=1}^k a_i b_i + a_{k+1}^2 b_{k+1}^2 \stackrel{(4)}{=} \quad (6)$$

$$\stackrel{(4)}{=} \left(\sum_{i=1}^k a_i^2 \right) \left(\sum_{i=1}^k b_i^2 \right) - \sum_{1 \leq i < j \leq k} (a_i b_j - a_j b_i)^2 + 2a_{k+1} b_{k+1} \sum_{i=1}^k a_i b_i + a_{k+1}^2 b_{k+1}^2 = \quad (7)$$

$$= \left(\sum_{i=1}^{k+1} a_i^2 - a_{k+1}^2 \right) \left(\sum_{i=1}^{k+1} b_i^2 - b_{k+1}^2 \right) - \quad (8)$$

$$- \sum_{1 \leq i < j \leq k} (a_i b_j - a_j b_i)^2 + 2a_{k+1} b_{k+1} \sum_{i=1}^k a_i b_i + a_{k+1}^2 b_{k+1}^2 =$$

$$= \left(\sum_{i=1}^{k+1} a_i^2 \right) \left(\sum_{i=1}^{k+1} b_i^2 \right) - b_{k+1}^2 \sum_{i=1}^{k+1} a_i^2 - a_{k+1}^2 \sum_{i=1}^{k+1} b_i^2 + a_{k+1}^2 b_{k+1}^2 - \quad (9)$$

$$- \sum_{1 \leq i < j \leq k} (a_i b_j - a_j b_i)^2 + 2a_{k+1} b_{k+1} \sum_{i=1}^k a_i b_i + a_{k+1}^2 b_{k+1}^2 =$$

$$= \left(\sum_{i=1}^{k+1} a_i^2 \right) \left(\sum_{i=1}^{k+1} b_i^2 \right) - \sum_{1 \leq i < j \leq k} (a_i b_j - a_j b_i)^2 - b_{k+1}^2 \sum_{i=1}^{k+1} a_i^2 - a_{k+1}^2 \sum_{i=1}^{k+1} b_i^2 + \quad (10)$$

$$+ 2a_{k+1} b_{k+1} \sum_{i=1}^k a_i b_i + 2a_{k+1}^2 b_{k+1}^2$$

Lemat 1.

T:.

$$\forall n \in \mathbb{N}_+ : \left\{ (i, j) \in \mathbb{N}_+^2 : 1 \leq i < j \leq n+1 \right\} - \left\{ (i, j) \in \mathbb{N}_+^2 : 1 \leq i < j \leq n \right\} = \quad (11)$$

$$= \left\{ (i, n+1) \in \mathbb{N}_+^2 : 1 \leq i \leq n \right\}$$

D:. Dla $n \in \mathbb{N}_+$, zdefiniujmy funkcję $A(n)$:

$$A(n) = \left\{ (i, j) \in \mathbb{N}_+^2 : 1 \leq i < j \leq n \right\} \quad (12)$$

$$\forall n : n \leq n+1 \implies \forall n : A(n) \subset A(n+1) \quad (13)$$

Dla każdego n i dla każdego i , dla którego $1 \leq i \leq n+1$:

$$(i, j) \in A(n) \iff i+1 \leq j \leq n \quad (14)$$

$$(i, j) \in A(n+1) \iff i+1 \leq j \leq n+1 \quad (15)$$

Tak więc

$$(i, j) \notin A(n) \wedge (i, j) \in A(n+1) \iff j = n+1 \quad (16)$$

Jeśli $j = n+1$, to

$$1 \leq i < j \leq n+1 \iff 1 \leq i \leq n \quad (17)$$

Co daje nam

$$A(n+1) - A(n) = \left\{ (i, n+1) \in \mathbb{N}_+^2 : 1 \leq i \leq n \right\} \quad \square \quad (18)$$

Korzystając z Lematu 1:

$$\left(\sum_{i=1}^{k+1} a_i^2\right) \left(\sum_{i=1}^{k+1} b_i^2\right) - \sum_{1 \leq i < j \leq k} (a_i b_j - a_j b_i)^2 - b_{k+1}^2 \sum_{i=1}^{k+1} a_i^2 - a_{k+1}^2 \sum_{i=1}^{k+1} b_i^2 + \quad (19)$$

$$+ 2a_{k+1}b_{k+1} \sum_{i=1}^k a_i b_i + 2a_{k+1}^2 b_{k+1}^2 \stackrel{L1}{=} \stackrel{L1}{=} \left(\sum_{i=1}^{k+1} a_i^2\right) \left(\sum_{i=1}^{k+1} b_i^2\right) - \left(\sum_{1 \leq i < j \leq k+1} (a_i b_j - a_j b_i)^2 - \sum_{i=1}^k (a_i b_{k+1} - a_{k+1} b_i)^2\right) - \quad (20)$$

$$- b_{k+1}^2 \sum_{i=1}^{k+1} a_i^2 - a_{k+1}^2 \sum_{i=1}^{k+1} b_i^2 + 2a_{k+1}b_{k+1} \sum_{i=1}^k a_i b_i + 2a_{k+1}^2 b_{k+1}^2 = \quad (21)$$

$$= \left(\sum_{i=1}^{k+1} a_i^2\right) \left(\sum_{i=1}^{k+1} b_i^2\right) - \sum_{1 \leq i < j \leq k+1} (a_i b_j - a_j b_i)^2 + \sum_{i=1}^k (a_i b_{k+1} - a_{k+1} b_i)^2 -$$

$$- b_{k+1}^2 \sum_{i=1}^{k+1} a_i^2 - a_{k+1}^2 \sum_{i=1}^{k+1} b_i^2 + 2a_{k+1}b_{k+1} \sum_{i=1}^k a_i b_i + 2a_{k+1}^2 b_{k+1}^2 =$$

$$= \beta(k+1) + \sum_{i=1}^k (a_i^2 b_{k+1}^2 - 2a_i b_{k+1} a_{k+1} b_i + a_{k+1}^2 b_i^2) - \quad (22)$$

$$- b_{k+1}^2 \sum_{i=1}^{k+1} a_i^2 - a_{k+1}^2 \sum_{i=1}^{k+1} b_i^2 + 2a_{k+1}b_{k+1} \sum_{i=1}^k a_i b_i + 2a_{k+1}^2 b_{k+1}^2 =$$

$$= \beta(k+1) + b_{k+1}^2 \sum_{i=1}^k a_i^2 - 2a_{k+1}b_{k+1} \sum_{i=1}^k a_i b_i + a_{k+1}^2 \sum_{i=1}^k b_i^2 - \quad (23)$$

$$- b_{k+1}^2 \sum_{i=1}^{k+1} a_i^2 - a_{k+1}^2 \sum_{i=1}^{k+1} b_i^2 + 2a_{k+1}b_{k+1} \sum_{i=1}^k a_i b_i + 2a_{k+1}^2 b_{k+1}^2 =$$

$$= \beta(k+1) + b_{k+1}^2 \sum_{i=1}^k a_i^2 + a_{k+1}^2 \sum_{i=1}^k b_i^2 - \quad (24)$$

$$- b_{k+1}^2 \left(\sum_{i=1}^k a_i^2 + a_{k+1}^2\right) - a_{k+1}^2 \left(\sum_{i=1}^k b_i^2 + b_{k+1}^2\right) +$$

$$+ 2a_{k+1}^2 b_{k+1}^2 =$$

$$= \beta(k+1) + b_{k+1}^2 \sum_{i=1}^k a_i^2 + a_{k+1}^2 \sum_{i=1}^k b_i^2 -$$

$$- b_{k+1}^2 \sum_{i=1}^k a_i^2 - a_{k+1}^2 \sum_{i=1}^k b_i^2 - 2a_{k+1}^2 b_{k+1}^2 + 2a_{k+1}^2 b_{k+1}^2 =$$

$$= \beta(k+1) \quad \square \quad (25)$$